

DEWDRIVE · DESIGN DECISION RECORD

DewDrive TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D8; item O1 remains proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation or a single proposed choice is therefore decided in favor of it. The one item without a recommendation stays open. TRL 4 work is on hold by the same instruction.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (TRL 2 session) and in DWD-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Budget	Raise budget_usd from \$400 to \$500 for a research prototype with fan and logging, rather than a fanless \$400 version or cost cuts that cost yield. Decided by Amish, 2026-09-25: go with recommendation.
D2	Problem wording	Reword the project.yaml problem line to: "Arid households without surface water or groundwater within reach depend on carried or trucked water; the air above them is an untapped supplementary source." Decided by Amish, 2026-09-25: go with recommendation.
D3	Sorbent	Silica gel with about 33 wt % CaCl ₂ , rather than LiCl composites, zeolite or MOF. Decided by Amish, 2026-09-25: go with recommendation.
D4	Collector layout	Single-stage flat glazed box with the condenser as its shaded floor, rather than dual-stage or a separate condenser. Decided by Amish, 2026-09-25: go with recommendation.
D5	Night airflow and power	Fan-assisted night airflow with a 10 W PV panel and a 12.8 V LiFePO ₄ battery, rather than natural airflow. Decided by Amish, 2026-09-25: go with recommendation.
D6	Flaps	Manual flaps, rather than automatic actuators. Decided by Amish, 2026-09-25: go with recommendation.
D7	Tilt and glazing	Fixed 20° tilt facing the equator; twin-wall polycarbonate glazing rather than glass. Decided by Amish, 2026-09-25: go with recommendation.
D8	Design point	20 °C and 40 % RH at night, 6.0 kWh/m ² of sun per day on the aperture; low-humidity case 25 % RH. Decided by Amish, 2026-09-25: go with recommendation.

Notes on the decided items:

- **D1.** `budget_usd` in `project.yaml` is now 500. R11 in DWD-REQ-001 v0.3 is redefined to \$500. The TRL 3 BOM totals \$488 (DWD-CAL-001, J1).
- **D2.** The problem line is applied to `project.yaml` and `README.md`. The review recommended no change to the `pitch`, which still describes the device accurately, so it is kept.
- **D3.** The salt loading is decided at about 33 wt %. DWD-CAL-001 (section E) found that this loading lets the solution fill the pores above 51 % RH, and proposes about 25 wt % instead; that is a new item awaiting Amish, not a change to D3.
- **D4 and D5.** DWD-CAL-001 (section B) found that air flowing past the trays captures only about 11 % of the vapour it carries, so R1 and R2 are not met as drawn. Drawing the air through the bed is proposed as a new item awaiting Amish; it keeps D4 and D5.
- **D8.** DWD-CAL-001 (C1) shows that the 6.0 kWh/m² day puts 4.24 kWh/m², not 4.5, in the 09:00 to 15:00 window; DWD-REQ-001 v0.3 carries the corrected figure.
- **SwapCell.** DewDrive uses its own 76.8 Wh battery and does not use a SwapCell pack. The portfolio decisions of 2026-09-25 on the SwapCell interface (v0.3 items: wake without CAN, charge-while-discharging mode, latch vibration rating) and on pricing shared packs once therefore do not change the DewDrive design or BOM.

Table 2. Items that remain open (Proposed, awaiting Amish).

#	Item	Why it stays open
O1	First partner and region for co-design, which will also supply the site climate data	No recommendation was made; the portfolio decision is to pick co-design partners per area later
O2	New TRL 3 proposals: through-flow trays, salt loading, radiation between bed and condenser, humid-night protection	Raised by DWD-CAL-001 after the decision; see <code>docs/REVIEW.md</code> , session 2026-09-25, TRL 3

■ Consequences

- DWD-PRB-001, DWD-PRC-001 and DWD-REQ-001 are revised to v0.3 to show the decisions: the key design choices of the precis are no longer "proposed", R11 carries the new budget, and the design point is decided.
- `project.yaml` carries `budget_usd: 500` and the reworded problem line.
- The partner question (O1) stays open in DWD-PRB-001.